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MPPT based on a novel load segmentations structure for PV applications

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ABSTRACT

The main disadvantage of photovoltaic (PV) systems is the mismatch between load and generation profiles. To overcome this disadvantage, it is either connected to battery bank or connected to the utility grid via inverters. While the battery choice is expensive and produces contaminating elements, the grid-connected mode is the most common choice. However, the inverters represent high percentage of the system cost that affect its potential to be a competitive economic replacement for other conventional energy sources. In addition, interfacing different users to the grid using inverters increase the total harmonics distortion (THD) and reduce the power quality of the grid. Moreover, exporting surplus energy generated from PV during off-peak periods with lower tariff reduces the economic effectiveness of PV systems. This paper proposes a segmented structure of the electric thermal loads with control algorithm that can achieve the MPPT objectives with no need for inverters. Moreover, with the right selection of PV size, it is possible to be locally consumed each generated kWh of energy without the need to export any to the main grid with low tariff. The segments of the electric thermal load that satisfy the MPP condition are connected to the PV while the rest segments are connected to main grid via set of electromagnetic relays. A simulation of the proposed structure and control algorithm was performed. The results shows that accurate MPPT was achieved according to the number of load segments. To increase the accuracy, the number of load segments should be increased.

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1. Introduction

Following the approach of using clean and eco-friendly energy sources; renewable energy sources began to rise; sources such as wind, photovoltaic, and hydropower now dominate the renewable energy market. These sources proved their capability to cover the increasing demand for electricity in societies. Photovoltaic technologies are the most common type of renewable energy source this is because it has a wider range of applications either domestic or commercial in addition to having multiple installation techniques such as ground or roof mounted it also comes in last place when talking about maintenance costs [1]. The inverter has been implemented to satisfy power specs requirements of loads and grid (if it is in connected mode). Inverters represent a significant percentage of the overall cost of the system. In addition, they increase energy loss due to their efficiency [2].

PV systems are most preferred due to their low cost and flexibility compared to wind turbines. They are the best option in large cities and towns where they can be easily deployed on top of residential and commercial buildings' roofs and can be integrated into other building structures such as windows and walls with lower efficiency [3]. The generated power of these systems can be easily integrated to the grid so the surplus energy should be managed and controlled as shown in [4].

Latterly, the techno-economic status of renewable energy power systems in residential community applications was discussed [5]. Reduction of PV system cost increases its potential to compete with other renewables. The cost of PV can be divided into the following categories: PV modules, inverters, wiring and safety, and mechanical framing and insulation [6]. The inverter cost 0.030 to 0.038 €/Wac for 100 to 200 kWac inverters. The starting price of PV modules in 2019 is 0.197 €/Wp. Taking into account that inverter lifetime is assumed to be half of the PV module lifetime. This means that the inverter must be replaced once during the lifetime

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of the system. The cost of replacement is taken into account in the calculation of the LCOE, separately from the OPEX. Impact of weighted average cost of capital, capital expenditure [7]. Hence, the inverters represent more than 30 % of the cost of the PV systems. The inverters allow PV systems to be connected to the utility grid. Hence, electrical energy can be exported during generation peak hours or imported during night and cloud coverage hours. Taking into consideration the difference in importing/exporting prices of electrical energy from/to utility grid, the Levelised cost of electricity (LCOE) generated kWh of electrical energy using PV system becomes high.

The deployment of each generated kWh from the PV system by local loads will effectively reduce the Levelised cost of electricity (LCOE) of generated PV kWh. Hence, a PV system would be an economically effective replacement for conventional energy sources [8].

To harvest maximum power from the PV system, different Maximum Power Point Tracking (MPPT) algorithms are used. MPPT is a method to find the maximum power point using PV power characteristics and an input voltage graph (P-V curve). MPPT tracker systems are connected to the DC-DC converters to be used to control the amount of output voltage on solar panel terminals to force the solar panel to generate the maximum power at various levels of irradiance. Nevertheless, sometimes it demonstrates seriously nonlinear multi-peak characteristics under shading or partial shading conditions.

Several MPPT were developed and implemented such as Perturb and observe (P&O), Advanced Perturb and observe (APO) [9], Current scanning method (CS), and Incremental Conductance technique (IC) [10]. It should be noted that all MPPT algorithms intended for the generation side only ignore the possibility to achieve MPPT using the load side. One of the criteria for MPPT classification is in terms of the control theory and the implantation method. There are three main types of the MPPT technique that can be divided into the following:

1. Traditional MPPT techniques.
2. MPPT Based on smart control.
3. MPPT under Partial shading conditions.

The traditional MPPT is mainly divided into two main types which are the Control methods on Parameter selection and the direct control based on the sampling data [11,12]. Utilizing the Sine-Cosine Algorithm (SCA) with adaptive control is analyzed to achieve accurate MPPT [13]. The MPPT Based on smart control is done by changing several parameters such as temperature and solar irradiation. Different artificial techniques are used to determine the MPPT as Fuzzy logic [14,15], neural network [16], adaptive PSO (APSO) [17] and sliding mode control [18]. The third type is the MPPT under partial shading conditions. To overcome the variation in the output power that results from the partial shading conditions, the researchers have developed an algorithm to increase the accuracy of the MPPT and the outpour consequently. Although the efficiency of this technique but it has problems such as requiring real-time data measurement [19] and lengthy calculations [20].

With the expansion of installing PV systems with inverters into the grid or standalone systems. Many issues have appeared when the inverter keeps operating at the MPPT mode such as overloading of the grid at peak power generation reported in [21] and high-temperature peaks and variations on switching devices due to the intermittency, which accelerates the degradation of the switching devices [22]. So, the separation between the inverter and the thermal load is highly recommended.

As shown from the previous discussion, all attention is focused on the generation side to achieve MPP with no effort recorded to adjust the load side. In addition, a common assumption that all loads are AC and can't be supplied directly from PV systems is evident. As a result, complicated and high-cost hardware is used to interface the PV systems to different types of loads. Whereas, most thermal loads can accept any form of electrical power (AC or DC). In addition, most of the electronic loads require DC supply and use AC to DC converters to be supplied from AC sources.

This paper represents a novel load segmentation MPPT (LS-MPPT). The technique performs MPPT by adjusting the load side rather than the generator side, hence, allowing load connection to PV with no need for inverters. Removing the inverters and additional hardware in grid-connected mode eliminates the total harmonics distortion from the grid. Moreover, the removal of these hardware devices will effectively decrease the overall system cost. The proposed structure is suitable for thermal loads at which the thermal coil can be supplied for either DC sources (PV) or AC sources (Grid). The proposed technique assumes the split of the heater coil into segments. The proper segments of the load are connected to the PV according to its MPP for a given irradiance.

If the thermal load is critical (it is desired to reach a given thermal power level) the rest segments of the heater coil are connected to the utility grid. Otherwise, the rest segments are lifted unconnected and the electrical thermal load will be 100 % reliable on PV.

The advantages of the proposed algorithm are as follows:

- Reduce the initial cost of PV systems by eliminating inverters and other hardware required to connect to the grid.
- Avoid interconnection approval and licensing when connecting PV system to grid.
- Reduce the undesired effects of connecting PV systems to the grid such as THD and the lack of generational inertia.
- Avoid exporting excess energy to the grid with low tariffs.
- For applications of hot water supply, hot water tanks can be a good energy storage device to compensate between the generation profile and consumption profile.

The paper is organized as follows: Section 2 discusses the proposed system structure; Section 3 introduces the control algorithm. Section 4 presents the model configuration used for simulations. Section 5 presents simulation results and discussion. Finally, Section 6 represents the conclusion and future work.

2. System structure

The proposed methodology of MPPT is based on load modification such that the load is split into multiple segments. The segments of the load that satisfy the MPPT condition are switched to the PV source while the other segments can be left unconnected or connected to the AC supplied from the main grid. Fig. 1 shows the hardware connection of both PV and grid sources to the load segments.

The loads that are suitable for such configuration are the ones that can operate effectively using DC or AC sources. Thermal loads, such as water heaters, boilers, ovens, stoves, and chillers are an example of loads that operate on both sources without any additional hardware. The only hardware requirement is to produce them with segmented heating coils with different values. Each segment is connected to the terminal supply (PV or Grid) via two-way control relay (or solid-state switches). Based on the control signals produced by the MPPT controller, the relay switches load segments to PV or grid. Hence, maximum power can be harvested from PV without the use of inverters. Moreover, the switches introduce an easy technique to control the thermal load as required regardless of the PV connection.

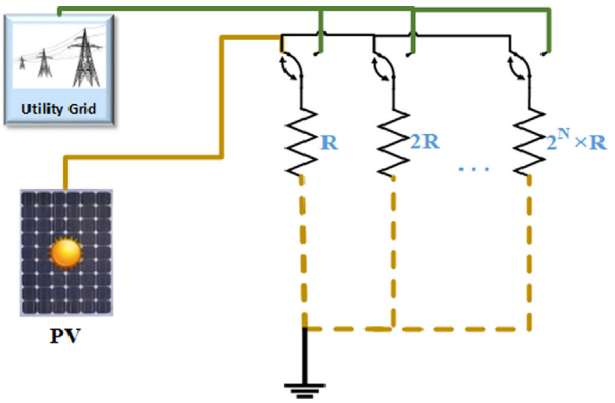


Fig. 1. LS-MPPT system configuration.

However, one may argue that the switching technique is the same as the one used in the inverters. In fact, the switching frequency for inverters ranges from 20 kHz to 2 MHz where stress and heat losses reduce the efficiency and lifetime of the switches. The higher the current rating of solid-state switches, the higher the price. The MPPT using segmented loads requires much cheaper switches that have much less switching frequency (the order of a few Hz).

The only drawback of the proposed structure is the incapability to export surplus energy generated by PV. However, the effectiveness of exporting surplus energy to the grid is in doubt due to two factors:

1. The exporting tariff is very low which may not compensate for the high initial cost of the PV system. The surplus energy of PV is frequently generated at the off-peak time at which it is useless.
2. The proper selection of PV size to accommodate the local required loads, especially for commercial and industrial loads, would be very effective.

The proposed load segments configuration is achieved by connecting segments of the load in parallel. Each segment has a value given by the equation.

$$R_{s_i} = 2^{-i}R \tag{1}$$

Where i is the index of the segment that ranges from 0 to N-1 and N is the number of load segments. The value R is given by:

$$R = R_L \sum_{i=0}^{N-1} 2^i \tag{2}$$

Where R_L is the total load resistance. Hence when connecting all load segments in parallel, the total resistance would be the original load resistance R_L. This configuration provides relative accuracy of 2^{-N}. The switches used to connect the load segment, either to the PV or to the grid, are controlled by the binary code representing the desired value of total resistance connected to the grid to achieve MPP.

This design allows two modes of operation: critical load mode and non-critical load mode. For critical load mode, the load segments should always be connected either to the PV or to the grid. Hence, the power consumption is controlled by the load. For non-critical load mode, only the load segments that satisfy the MPP are connected to PV harvesting its maximum power while the reset segments are left unconnected. Hence, the power consumption is controlled by the available power from PV.

As shown in Fig. 2, the configuration with N load segments provides $2^N - 1$ different load lines with relative accuracy of 2^{-N}. The load line with slope $\frac{2^N-1}{R}$ is matched to the maximum irradiance curve to provide MPP. All other load lines are uniformly separated by the minimum slope step $(1/R)$.

3. Control algorithm

In order to track the time-varying maximum power point of the solar array depending on its operating conditions of irradiation and temperature, the MPPT control technique plays an important role in practical PV systems. A variety of MPPT schemes and several sensors-less approaches have been proposed in the literatures [23].

This paper proposes the load segmentation MPPT (LS-MMPT) control technique. The output power of PV arrays varies with weather conditions such as solar irradiation and atmospheric temperature. Therefore, real-time MPPT control for extracting maximum power from the PV panel becomes indispensable in PV generation systems [24,25].

The control algorithm starts by measuring PV current $I_{PV}(k)$ and terminal voltage $V_{PV}(k)$ at each sample k and use them as input to the controller. The power is calculated using the following formula:

$$P_{PV}(k) = V_{PV}(k) \times I_{PV}(k) \tag{3}$$

The operating point of the PV is given by the intersection of the load line and PV characteristic curve. The load line equation is given by:

$$V_{PV}(k) = \frac{I_{PV}(k)}{G(k)} \tag{4}$$

and I-V characteristic of the ideal photovoltaic cell is given by:

$$I_{PV}(k) = I_L - I_O\left[exp\left(\frac{V_{PV}(k)}{V_T}\right) - 1\right] \tag{5}$$

Where I_L is the current generated by the incident light (it is directly proportional to the sun irradiation), I_O is the leakage current, V_T is the thermal voltage, $I_{PV}(k)$ is the PV current at instance k, and $V_{PV}(k)$ is PV terminal voltage at instance k [26,27].

The slope of the current operating point at instant k on the power-voltage curve is given by:

$$E(k) = \frac{P_{PV}(k) - P_{PV}(k-1)}{V_{PV}(k) - V_{PV}(k-1)} \tag{6}$$

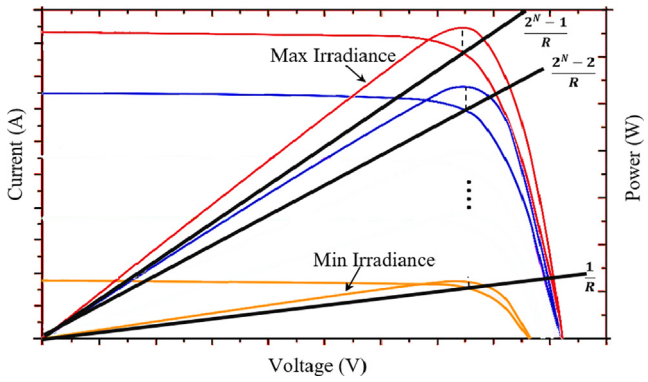


Fig. 2. The P-I curve of PV cell at different irradiance and different load curves based on the proposed structure.

The control objective is to adjust the load line slope ($G(k) = 1/R(k)$) (load admittance) such that $E(k) = 0$ (The point at which MPP is achieved has zero slope). Using equations (4) and (5), the power characteristic curve of the load and photovoltaic cell is given by the following respectively:

$$P_{PV}(k) = \frac{I_{PV}^2(k)}{G(k)} \tag{7}$$

$$P_{PV}(k) = \left(I_L - I_O\left[\exp\left(\frac{V_{PV}(k)}{V_T}\right) - 1\right]\right)V_{PV}(k) \tag{8}$$

Hence, to achieve the proper value of $G(k)$ that leads to MPP, it is required to solve equations (7) and (8) under the condition of zero slope of the power curve given by (6) ($E(k) = 0$). It is impossible to solve the equations from (6) to (8) to find the optimal $G(k)$ analytically due to nonlinearity and coupling between the variables. It is much easier to represent the problem graphically and find the solution numerically by iteration.

Assuming the case as shown in Fig. 3 at which the operating point at instant k is given by the intersection of load line that has slope $G(k)$ and any given photovoltaic cell curve for specific irradiance (shown in red line). The corresponding voltage and power is $V_{PV}(k)$ and $P_{PV}(k)$ respectively.

A positive slope (Fig. 3a) of the power curve at instant k represents a load value less than the optimal value at MPP. Hence, to improve the extracted power from the photovoltaic cell, the slope of the load line should be decreased (the load value should be increased). While the negative slope (Fig. 3b) of the power curve at instant k represents a load value greater than the optimal value at MPP. Similarly, to improve the extracted power from the photovoltaic cell, the slope of the load line should be increased (the load value should be decreased). Accordingly, the adaptation of the load line to approach the MPP is dependent on $E(k)$ as follows:

$$G(k+1) = G(k) - \mu E(k) \tag{9}$$

Where μ is a value that defines the iteration step size. Similarly, the second iteration is given by:

$$G(k+2) = G(k+1) - \mu E(k+1) \tag{10}$$

Substituting (9) in (10).

$$G(k+2) = G(k) - \mu E(k) - \mu E(k+1) = G(k) - \mu \sum_{i=0}^{1} E(k+i) \tag{11}$$

After m samples.

$$G(k+m) = G(k) - \mu \sum_{i=0}^{m-1} E(k+i) \tag{12}$$

$$G(k+m) = G(k) - G_E(k+m) \tag{13}$$

Where $G_E(k+m)$ is the change of load admittance after m iteration from the current sample k which is given by:

$$G_E(k+m) = \mu \sum_{i=0}^{m-1} E(k+i) \tag{14}$$

However, due to the segmentation of the load to discrete values, the adaptation of the admittance at instant k ($G_E(k)$) is limited to integer values:

$$G_e(k) = \frac{n_E(k)}{R} = \frac{1}{R} \times int\left(K_c \sum_{i=0}^{m-1} E(k)\right) \tag{15}$$

Where, K_c is the controller gain ($K_c = R \times \mu$), R is given by (2), $n_E(k)$ can take any value between 0 and N-1 and.

$$int(x) = \begin{cases} K+1 & if\ 0.5 \geq x - K \geq 1 \\ K & 0 \geq x - K > 0.5 \end{cases} \tag{16}$$

Rewriting equation (13) using integer multiples of the minimum load segment ($.1/R$)

$$\frac{n(k+m)}{R} = \frac{n(k)}{R} - \frac{n_E(k+m)}{R} \tag{17}$$

$$\therefore n(k+m) = n(k) - n_E(k+m) \tag{18}$$

However, $n(k+m)$ should never exceed the two limit values of connected load segments (non or N-1), hence.

$$n(k+m) = \begin{cases} 0 & n(k) - n_E(k+m) \leq 0 \\ n(k) - n_E(k+m) & 0 \leq n(k) - n_E(k+m) \leq N-1 \\ N-1 & n(k) - n_E(k+m) \geq N-1 \end{cases} \tag{19}$$

Equation (19) represent the overall adaptation mechanism at which the value $n(k)$ is converted to its binary form and used to switch the load segments On and Off to achieve the optimal load curve for the current MPP. The controller gain K_c is a design value

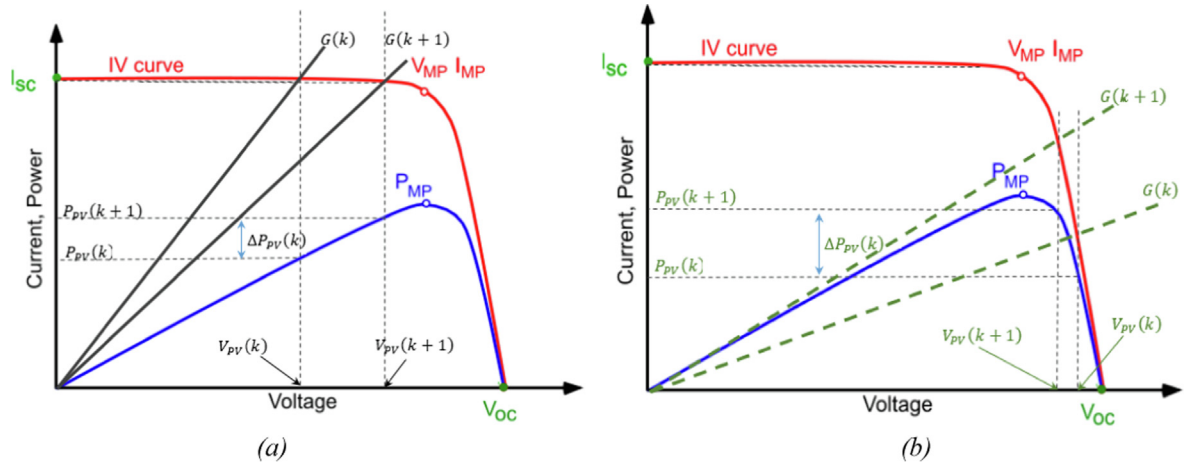


Fig. 3. Load line and photovoltaic cell curve mismatching MPP at instant k. (a) the current operating point to the left of the MPP and (b) the current operating point to the right of the MPP.

that should be selected carefully to achieve fast transient yet stable iteration (the slope should always converge to zero with time). The stability of the iteration mechanism will be discussed in section 5.

However, using the adaptation mechanism presented by (19) could lead to a sustained oscillation around the MPP. The oscillation frequency is dependent on the controller gain, the sampling rate, the current power operating point and the number of segments used for the load. The load segments at different connection arrangements produce $(2^N - 1)$ discrete load lines adjusted to predefined power-voltage curves to achieve MPP. However, at different irradiance values, the slope at the operating point will never be zero. The controller will produce positive and negative control action in high frequency switching between two adjacent load lines surrounding the current MPP. This will result in a high switching frequency to the load segment and add undesired stress to the switches.

To avoid the high-frequency switching trend, the controller should stop adaptation (connecting/disconnecting load segments) whenever the system achieves power close to the MPP as long as the control process will not improve the harvested power. Using a dead-band of $E(k)$ is ineffective as the slope of the P-V curve is almost constant over a wide range of PV characteristics.

If the irradiance level is known, then it is possible to identify the proper load line that will produce power close to MPP. However, PV voltage and current are the only quantities that are measurable. Fig. 4a shows how we can identify the irradiance level at the current operating point. The figure shows a power $P(k)$ at instant k and a different corresponding voltage $V_1(k)$, $V_2(k)$ and $V_3(k)$. Given that $P(k)$ is at $V_1(k)$ then, the irradiance should be identified by the red curve, and hence, the current operating point is too far from MPP. On the other hand, if $P(k)$ is at $V_2(k)$ or $V_3(k)$, the irradiance curve must be the blue one and the load line is close to MPP.

Based on the power-voltage relation presented by the PV characteristic curve, if a zone formed by two power limits and two voltage limits can be defined for each discrete irradiance curve at which only one discrete load line will satisfy, as much as possible, the MPP, then it is possible to switch the adaptation Off within this zone.

Fig. 4b shows how it is possible to identify the adaptation Off zone. The figure shows different load lines indexed by $i-1, i$, and $i+1$. Each load line is matching the MPP for its corresponding irradiance curve and produces power P_{i-1}, P_i, and P_{i+1} at voltage V_{i-1}, V_i, and , V_{i+1} respectively. Focusing on the load line i, the margin at which the load line can't be adapted is defined by hypothetical load lines $i-0.5$ and $i+0.5$ creating a yellow zone surrounding the load line as shown in Fig. 4b. The borders of the yellow zone will intersect with the j I-V curve of the current irradiance in $V_{i-0.5}$ and $V_{i+0.5}$ (vertical blue zone) producing power $P_{i-0.5}$ and $P_{i+0.5}$ respectively. Accordingly, If the PV is producing power ranging from $P_{i-0.5}$ to $P_{i+0.5}$ and the PV terminal voltage ranges from $V_{i-0.5}$ to $V_{i+0.5}$ (a zone presented by green rectangular in Fig. 4b, it can be concluded that the current load line i is adequate to achieve MPP and the adaptation mechanism will switch Off to avoid oscillation between either load line i and $i-1$ or i and i $+1$.

However, how to find $P_{i-0.5}, P_{i+0.5}, V_{i-0.5},$ and $V_{i+0.5}$? Because the load lines are equiangular lines, the hypothetical load-line $i-0.5$ will be at the midway between load line i and $i-1$, while the hypothetical load-line $i+0.5$ will be at the midway between load line i and $i+1$. The lines will intersect with the linear portion of I-V curve, hence the corresponding power and voltage will keep the uniformity. It can be noted that a minor nonlinearity is evident in the right portion of the curve, however, the figure presents an exaggeration of load lines spacing and nonlinearity can be neglected for simplicity in real case. Based on this analysis, the region at which the adaptation mechanism will be switched off is defined by:

$$\begin{aligned} P_{i-0.5} &= (P_{i-1} + P_i)/2 \\ P_{i+0.5} &= (P_{i+1} + P_i)/2 \\ V_{i-0.5} &= (V_{i-1} + V_i)/2 \\ V_{i+0.5} &= (V_{i+1} + V_i)/2 \end{aligned} \tag{20}$$

Adding those constrains to adaptation mechanism, equation (19) will be as follow.

$$n(k+m) = \begin{cases} 0 & n(k) - n_E(k+m) \le 0 \\ n(k) & \begin{array}{c} if\,(P_{i-1}+P_i)/2 < P_{PV}(k) < (P_{i+1}+P_i)/2 \\ and V_i + V_{i-1} < V_{PV}(k) < V_i + \Delta V_{i+1} \\ for i = 1 to N-1 \end{array} \\ n(k) - n_E(k+m) & otherwise \\ N-1 & n(k) - n_E(k+m) \ge N-1 \end{cases} \tag{21}$$

Hence, the load line will remain unchanged if the power and voltage are within ceiling values $((P_{i+1} + P_i)/2)$, $((V_i + \Delta V_{i+1})/2)$ respectively and power and voltage are within floor values

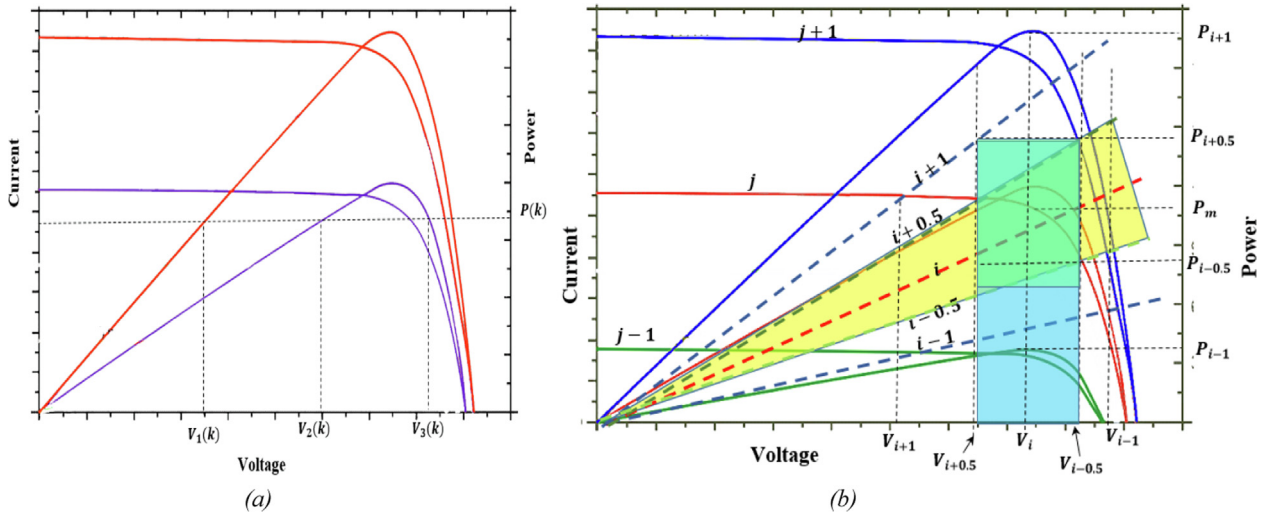


Fig. 4. Discrete Load lines and associated power and voltage limits.(a) relation between power level, voltage and irradiance curve (b) how power and voltage limits can be calculated to create adaptation OFF zone.

$((P_{m-1} + P_m)/2)$, $((V_i + \Delta V_{i-1})/2)$ respectively. The power and voltage ceiling and floor values represent the optimal MPP region, according to the current PV operating point, at which any modification to the number of connected load segments wouldn't improve the performance of harvested power.

Fig. 5 shows the flowchart of the algorithm used for the proposed LS-MPPT. The algorithm starts by measuring the PV voltage and power. Those values are used to determine whether the system is between the ceiling and floor values of the current operating power curve. If they are, no action is required, and the system is working close as maximum as possible to the MPP. Otherwise, the algorithm calculates the slope of the power curve and determines the error required to update the connected load segments, convert them to discrete values, and apply them to switch On and Off the load segments.

4. Model configuration

The system consists of 8 PV modules connected in series, a load of total resistance (R_L) 27.5 Ω and a set of 8 two-way relays. The load is segmented to 4 segments with values 51.56 Ω, 103.12 Ω, 206.24 Ω and 412.48 Ω (according to equation (1) and (2)). The load and the PV system are rated are at 1.59 kW and 210 V. The data of the system are listed in Table 1. The temperature was set to 25 °C and the irradiance was changed in steps 1000 W/m^2, 500 W/m^2, and 200 W/m^2.

5. Simulation results

The simulation was conducted using Matlab and Simulink environments and applying two control algorithms, the 1st is the LS-MPPT algorithm described by equations (19) and the 2nd is its enhanced version described by equations (21). The system is

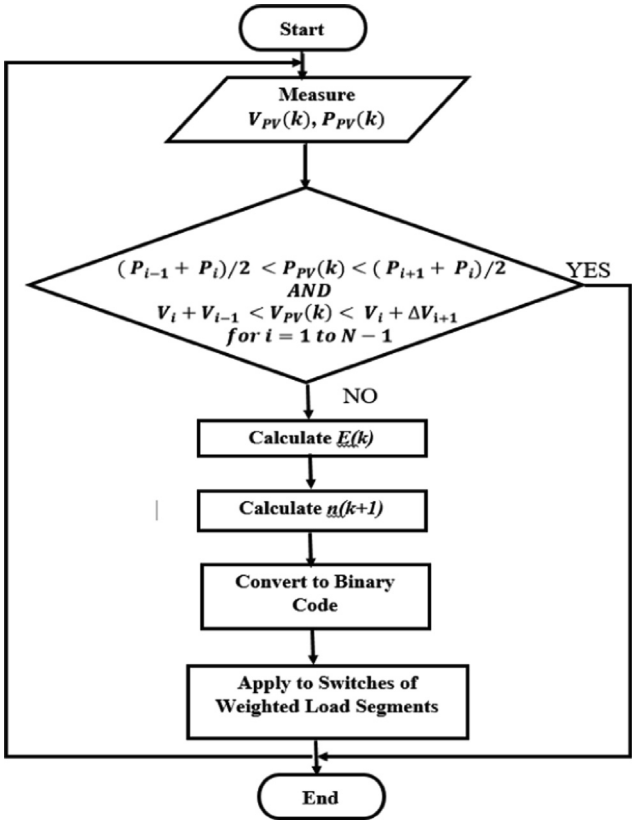


Fig. 5. Algorithm flowchart.

Table 1
System Data.

Symbol	Quantity	Values
P_{MPPT}	Rated power	199 W
V_{MPPT}	Rated voltage	26.3 V
I_{MPPT}	Rated current	7.6 A
V_{OC}	Open-circuit voltage	32.9 V
I_{SC}	Short-circuit current	8.21 A
N_S	Number of series cell	54
N_P	Number of parallel cells	1
N_{Sm}	Number of series module	8
N_{Pm}	Number of parallel modules	1
C	PV module capacitor	0.47μF
T_C	Atmospheric Temperature	25 °C
G_N	Solar Irradiation	1000 W/m^2
V_S	Grid voltage in RMS	220 V
F_{SM}	Sampling frequency	33 kHz
F	Line frequency	50 Hz
R_L	Load resistance	27.5 Ω
N	Load segments	4

subjected to 6 different irradiance levels for 6 s ([1000 W/m^2, 800 W/m^2 , 700 W/m^2, 500 W/m^2, 200 W/m^2 and 600 W/m^2). The PV temperature is kept unchanged as the effect of changing the temperature will have the same impact as changing the irradiance.

Fig. 6 shows the consumed power from PV and grid together with irradiance. It can be noted that the control algorithm switches On the proper segments of the loads to the PV and grid to satisfy the required power and achieves MPP. Whenever the irradiance is maximum (1000 W/m^2), all load segments are connected to the PV and no energy is consumed from grid. As the irradiance decreases (intervals from 0.5 *sec* to 2 *sec*), the controller switches Off the proper load segments from PV tracking MPP at a lower power level. Those unconnected load segments from PV are connected to grid increasing supplied power from the grid and maintaining the load power as desired. To examine the algorithm in positive change of irradiance, the irradiance increased from 200 W/m^2 to 600 W/m^2 then to 1000 W/m^2. The adaptation mechanism continue to adjust the connected load segment to PV to achieve MPP.

Fig. 7 shows the PV current and voltage. It can be noted that at transient when the irradiance changes, the current and voltage change accordingly. The control algorithm adjusts the load, and hence the current, sliding on the current–voltage PV characteristics towards the MPP either in the negative direction (when irradiance decreases) or in the positive direction (when irradiance increases). The voltage level track the value corresponding to MPP (around 210 V for all PV characteristic curves).

Fig. 8 shows the total value of the load segments connected to PV (upper curve) and the total value of the load segments connected to the grid (lower curve). It can be noted that at time intervals of 0.5:0.9 *sec*, 2.9:3.4 *sec*, and 4.8:5.3 *sec* the PV receives full irradiance (1000 W/m^2) the total load segments are connected to PV (27.5 Ω) and the grid is left open circuit ($\infty\Omega$). As the irradiance changes, the adaptation mechanism switches On and Off the load segments between PV and grid to adjust the total PV-connected load segments to satisfy the MPP condition. However, because values of the load segments are discrete, the adaptation mechanism switches between two adjacent values causing undesired oscillation. A focus is made on the interval from 1.75 *sec* to 2 *sec* to show the oscillation behavior of the adaptation algorithm at which the PV receives minimum irradiance (200 W/m^2). It can be noted that the algorithm switches load segments to PV to achieve a total value between 137.5 Ω (1st and 2nd load segments connected in parallel) and 103.125 Ω (3rd segment only). Accordingly, the unconnected segments to PV will be connected to the grid (as shown in the

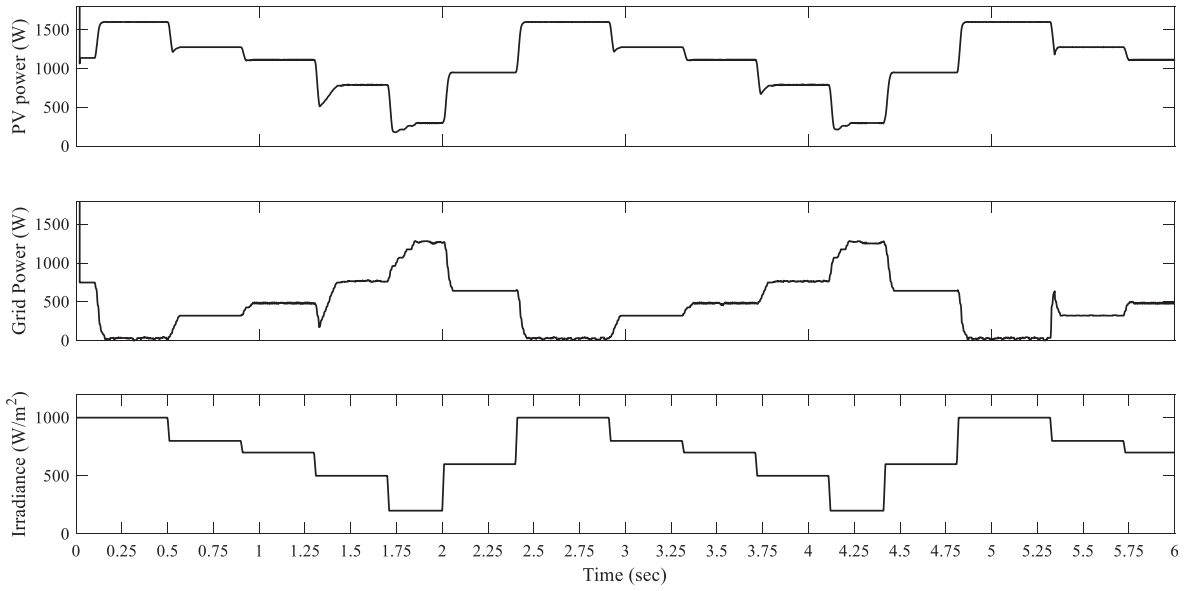


Fig. 6. The load power consumption from PV (upper curve) and from grid (The middle curve). The lower curve shows the irradiance. (LS-MPPT).

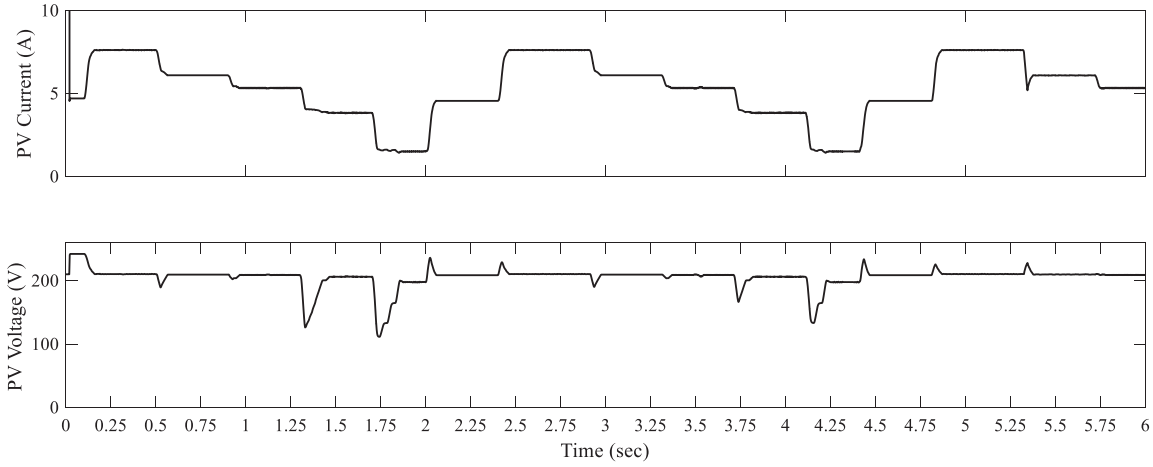


Fig. 7. PV current and voltage (LS-MPPT).

lower curve of the focus) and the grid-connected total load segments value will switch between 34.375 Ω (3rd and 4th segments are connected in parallel to the grid) and 37.5 Ω (1st, 2nd and 4th segments are connected in parallel to the grid).

Figures (9), (10), and (11) show the results using the enhanced LS-MPPT algorithm (applying equation (21)). It can be noted that the same power, voltage, and current characteristics of the 1st algorithm were achieved using the 2nd algorithm. However, Fig. 11 shows that the enhanced algorithm eliminates the switching oscillation totally during the simulation course with different irradiance levels.

Finally, Fig. 12 shows the ratio between the energy consumed by the load from PV and the grid. The average output power at rated values of a PV system is 1120 W according to the irradiance waveform shown in Figs. 6 and 9 (lower curve). In 5.9 s (simulation period starting from 0.1 *sec*), the total harvested energy from PV, using the proposed structure and algorithm, was 1.83 Wh and the total consumed energy from the grid was 0.78 Wh which represents a ratio of 70 % from PV and 30 % from the grid as shown in Fig. 12. To investigate the efficiency of the proposed structure and algorithm, the actual output energy is compared to the rated value at the same irradiance condition. In addition, for comparison purposes, the MPPT given in [26] is implemented to the same load and irradiance condition. The results of energy and its percentage (referred to rated PV system output energy) are shown in Table 2. The efficiency of the proposed LS-MPPT is very close to the conventional MPPT. For sure it is lower due to the quantized value of the load segments that can't achieve perfect MPPT as discussed in section 3. Fig. 10.

As stated in section 3, the system is highly nonlinear and the study of the algorithm stability analytically is impossible. However, algorithm stability can be determined using phase plane at which, convergence to origin proves system stability. Fig. 13 shows the phase plane for different transient instants (0.1sec, 2.0 *sec*, and 2.9 *sec*). The selected instants reflect positive and negative transition with different magnitude. Arrows were added to the figure to enable ease recognition of the direction of phase plane trajectory with time. The figure shows that all trajectories converge to origin

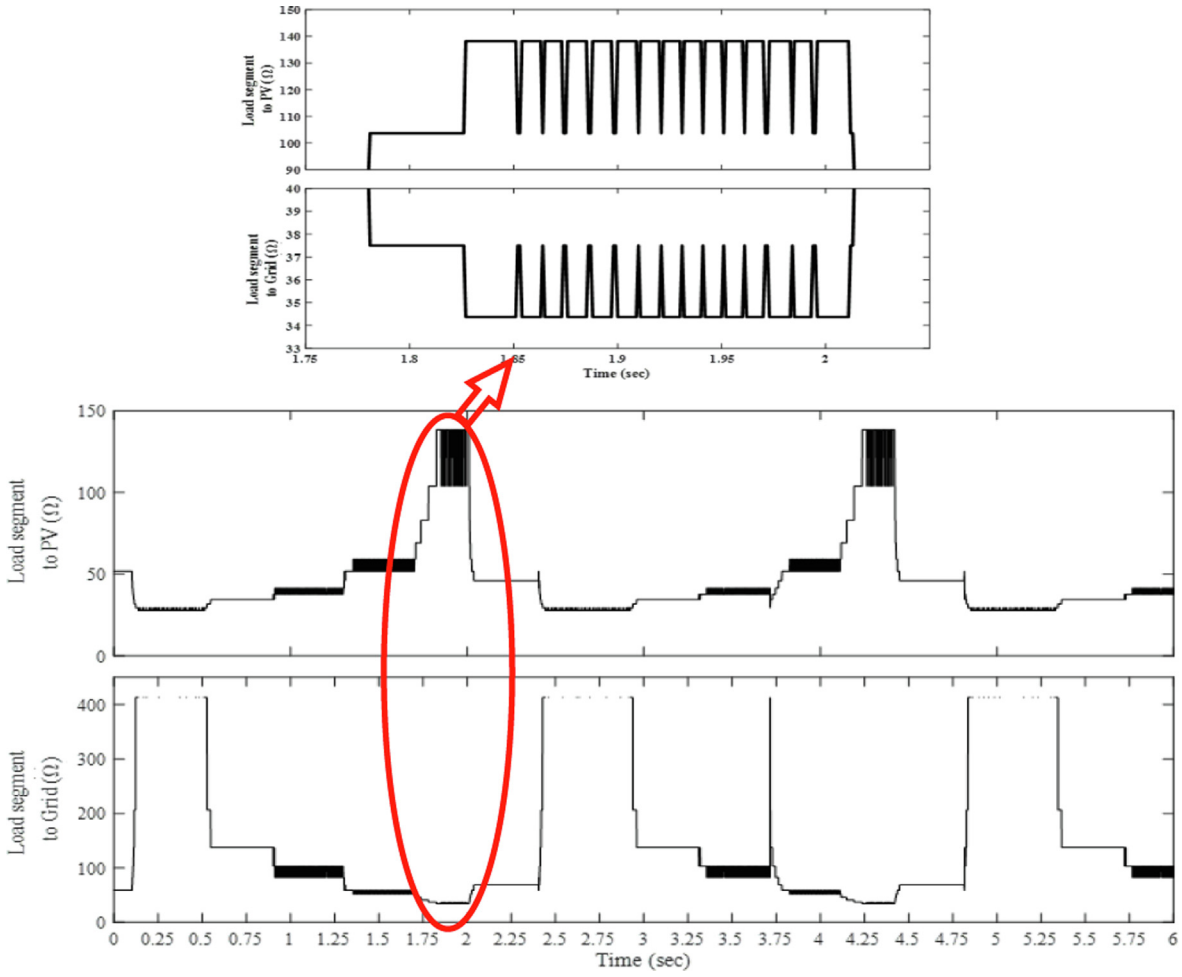


Fig. 8. PV load value connected to PV (upper curve) and grid (lower curve). (LS-MPPT).

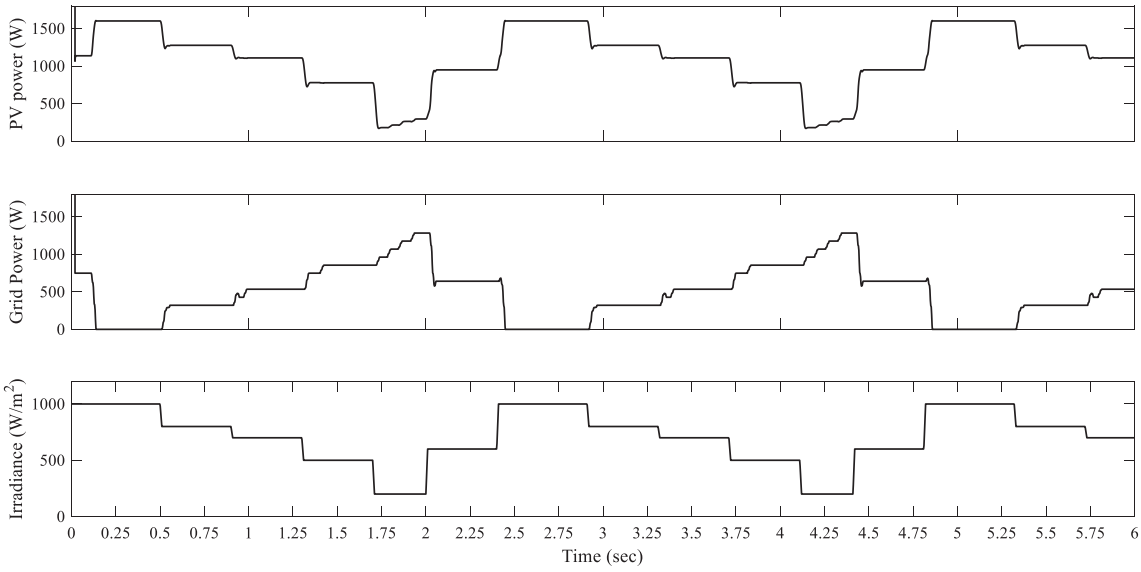


Fig. 9. The load power consumption from PV (upper curve) and from grid (The middle curve). The lower curve shows the irradiance. (Enhance LS-MPPT).

Table 2
Comparison between Proposed LS-MPPT and conventional MPPT.

	Energy (Wh)	Percentage
Rated PV system	1.866	100.00 %
Proposed LS-MPPT	1.83	98.07 %
Conventional MPPT [26]	1.86	99.68 %
Grid	0.78	41.80 %

Note: Percentage Normalized to rated PV system.

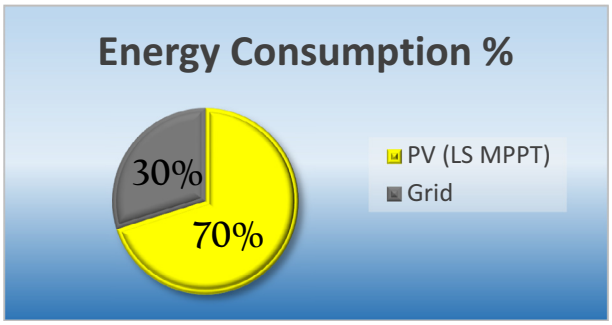


Fig. 12. Percentage of energy consumption by the load from PV and grid.

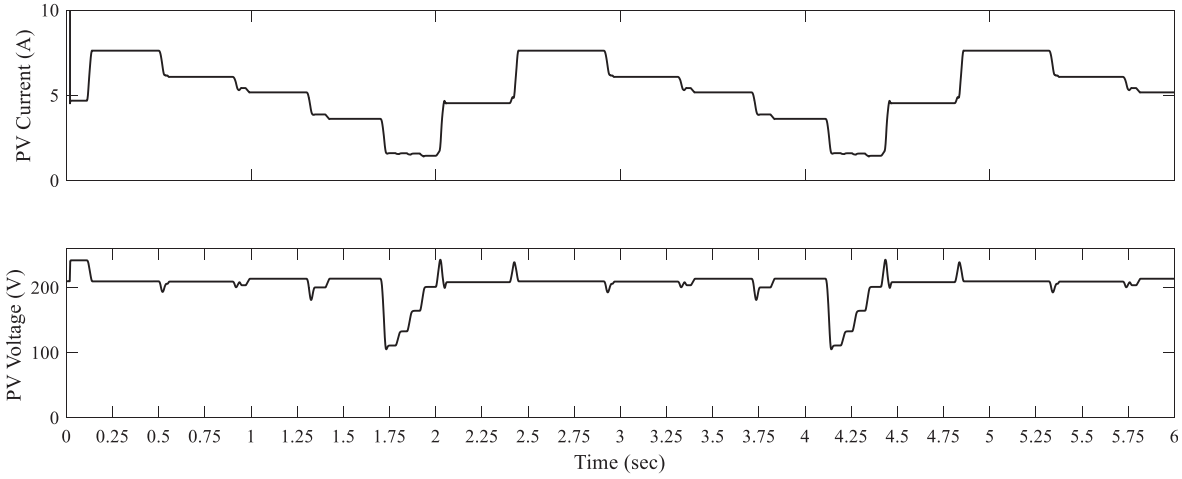


Fig. 10. PV current and voltage (Enhanced LS-MPPT).

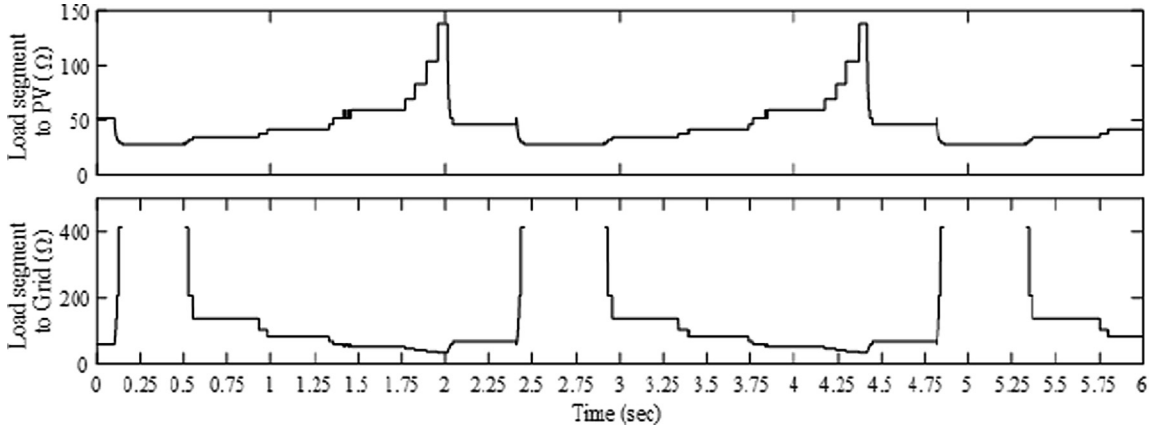


Fig. 11. PV load value connected to PV (upper curve) and grid (lower curve). (Enhanced LS-MPPT).

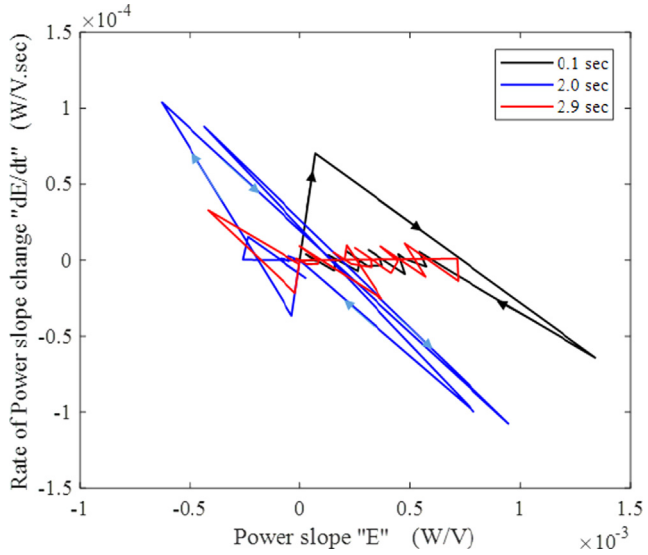


Fig. 13. Phase plane for proposed LS-MPPT at different transient instants (0.1sec, 2.0 *sec*, and 2.9 *sec*).

which confirms the system stability at different transient conditions.

6. Conclusion

In this paper, a novel segmented structure of the electric thermal loads with a control algorithm that can achieve the MPPT objectives is introduced. The structure allows the operation of the PV system at MPP with no need for inverters which would reduce the cost and increase durability. Furthermore, the removal of the power electronics devices from the on-grid PV system eliminates the total harmonics distortion added by those devices. Moreover, the structure and control algorithm allows local consumption of each generated kWh of energy without the need to export any to the main grid with low tariffs.

The structure of the system was presented. At this structure, the load is segmented to binary-weighted segments connected in parallel to PV or grid through switches. A control algorithm, based on the feedback of the slope of the power curve, is used to switch On and Off the load segments to PV or grid. The control algorithm achieves MPP at zero slope of the power curve. Moreover, an enhancement to the proposed control algorithm was introduced to avoid switching oscillation due to the quantization of load segments. The enhancement technique assumes an "adaptation off zone" with power and voltage ceiling and floor values at which the adaptation mechanism will be switched off.

A simulation was conducted to validate the proposed algorithm and its enhancement version. The results show that proposed structure and control algorithm were able to harvest 98 % of energy from PV during the simulation period in presence of a dramatic change of irradiance. The algorithm was able to satisfy the load demand from both PV system (70 %) and grid (30 %). In addition, the enhanced control algorithm eliminate the oscillation that may be experienced due to the quantization of load segments. Hence, the switching stress is released while the performance is kept as desired. The algorithm stability was examined using the phase plane for different transient conditions and shows conversion to the origin, and hence stability, at each transient condition.

The system advantages are listed in the introduction section which opens research on a wide range of applications. One of the most effective applications (to our point of view) is a hot water supply system in which, electrical heater coils can accept both AC and DC supply and it includes insulation tanks that can compensate for the generation and demand profiles. However, some can argue the presence of the following disadvantages:

1. Despite the high percentage of harvested energy from PV, it is always less than the perfect MPPT. This is because only discrete values of the load are available and will never match the exact MPP. However, the tradeoff is high with the potential to reduce the PV system's initial cost. In addition, the elimination of inverters from the system eliminates one device that is a source of power loss that may overcome the loss introduced by the proposed algorithm.
2. The excessive generated energy by PV using this algorithm will be lost as it can't be exported to the grid. However, the proper sizing of PV and the selection of applications in which a type of energy storage device exists (such as a hot water insulation tank), excessive energy can be minimized and all generated energy can be deployed.

The introduced work opens a window to a variety of applications and requires more investigation of its economic efficiency. This paper focused on the dynamic model and validity of the proposed structure and algorithm. However, an economic comparison with similar PV systems that use inverters or other similar renewable energy sources, such as solar heater systems, is a high potential for future work. Moreover, the effect of increasing the number of segments, the optimal segment number, and the application of other control methodology to the proposed structure can be addressed in future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A:. Circuit diagram

See Figs. 14-16.

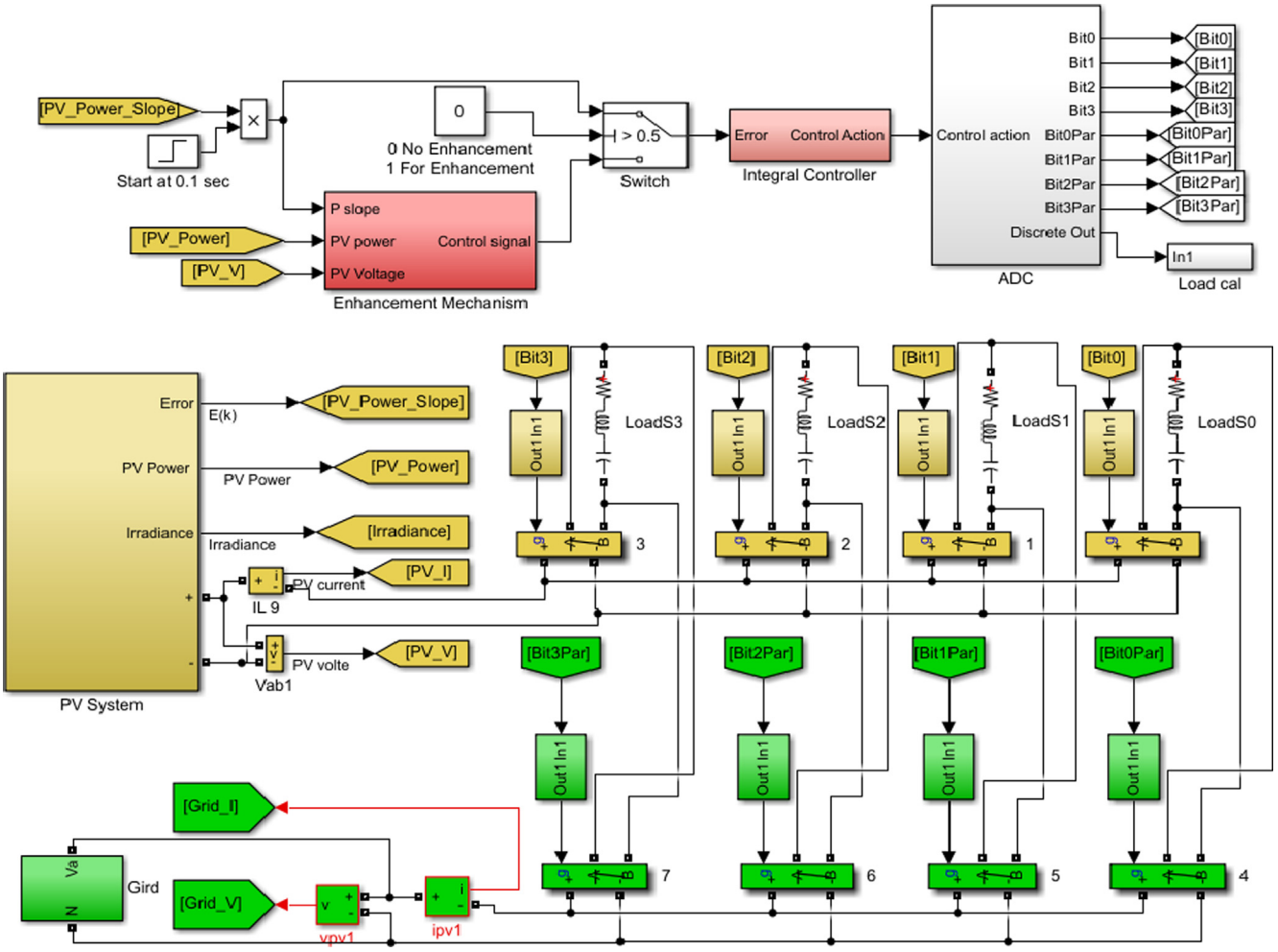


Fig. 14. LS-MPPT circuit diagram.

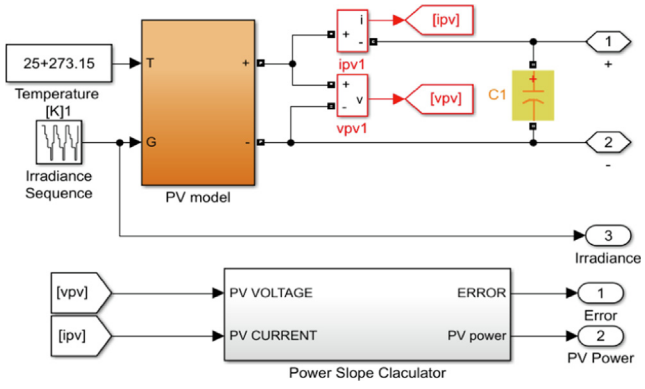


Fig. 15. PV system detailed circuit diagram.

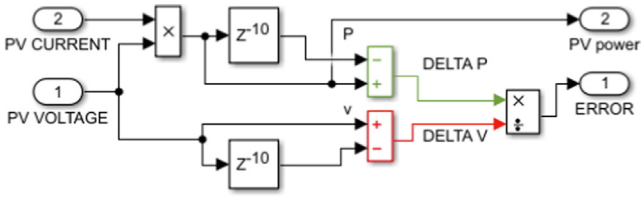


Fig. 16. PV power slope calculator.

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